NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER.

NAMES: _____ DATE OF BIRTH: _____

SCHOOL/INSTITUTION: _____

CLASS: _____ GENDER: _____ DISTRICT: _____

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

Do not turn this page until you have been instructed to do so. This competition is conducted on the assumption that proper security is maintained. Uganda Mathematical Society reserves the right, should there be evidence of any breach of security or any suspicion, to reject scripts from a particular school/institution.

PLEASE, NOTE CAREFULLY:

1. **WRITE YOUR ANSWERS IN THE SPACES PROVIDED. MORE SPACE IS AT THE END.**
2. Date: Saturday, 28th May, 2022.
3. Time: Exactly 2 hours and 15 Minutes (from 9 : 00am to 11 : 15am)
4. Type:
   (a) This paper consists of two sections and each section carries 50 marks.
   (b) Attempt as many questions as you can. The total marks scored from all the questions you attempt will be your final score.
   (c) No marks will be awarded for an answer if NO clear and logical layout of the working is shown.
5. **Indicate your names, school and district on all your answer sheets in capital letters.**
6. All diagrams/figures in this contest are not to scale.
7. Immediately the paper is done, the answer script(s) must be forwarded to:

   David Ddumba Walakira,
   Contest Coordinator, Uganda Mathematical Society,
   c/o Department of Mathematics, Makerere University,
   P.O. Box 7062, Kampala.
   Telephone contact: $+256-414-540692$.

   Answer scripts should be received **latest, Monday** 30**th May,** 2022.
8. *ALL participants who score a pass mark are cordially invited to the certificate and prize giving ceremony on Saturday* 23*rd July,* 2022 *at 8:00am at Makerere University, Kampala.*
9. **UGANDA MATHEMATICAL SOCIETY** wishes you success in this year's contest.

SECTION A : 5 marks each

1. The size of one angle of an isosceles triangle is 50 degrees. The size of the other angles (in degrees) in this triangle could be (circle the right answer)

   A. $65, 25$ B. $60, 70$ C. $65, 80$ D. $50, 80$

2. What time is 145 minutes after $9:06$ a.m?

3. What is the median among $\frac{5}{6}, \frac{6}{7}, \frac{2}{5}, \frac{1}{3}, \frac{4}{5}, \frac{3}{7}$?

4. A, B, C, D and E play a game in which each is either a lion or a goat. A lions' statement is always false and a goats' statement is always true.

   A says B is not a goat

   C says D is a lion

   E says A is not a lion

   B says C is not a goat

   D says E and A are different kinds of animals.

   Who are the lions?

5. A rectangle has length x cm and width y cm. A triangle has base 18cm and height $x$cm. Both have the same area. What is the value of y?

6. Coxy owns more cats than Alice and more dogs than Bruno. Alice owns more dogs than Coxy and fewer cats than Bruno. Which of the statements must be true? (Circle the right answer)

   A. Bruno owns the fewest cats

   B. Coxy owns the most cats

   C. Alice owns the most dogs

   D. Coxy owns the fewest dogs

7. An operation is defined as $a * b = a^2 + 2^b$. What is the value of $-3 * -3$?

8. In an increasing list of consecutive whole numbers, the 3rd and 4th numbers in the list add to 11. What is the 2022nd number in the list?

9. In a sequence, each term after the first one is the sum of the squares of the digits of the previous term. For example if the first term is 42 then the next term is $4^2 + 2^2 = 20$. The next term is $2^2 + 0^2 = 4$, Next $4^2 = 16$, next $1^2 + 6^2 = 37$ and so on giving the sequence $42, 20, 4, 16, 37, \ldots$. If the first term is 44, what is the 2022nd term?

10. Viola picks any number from 1 to 10. Bally adds 5 to this number and calls his result b. Carol subtracts 5 from Viola's number and calls her result c. What is the value of $b - c$?

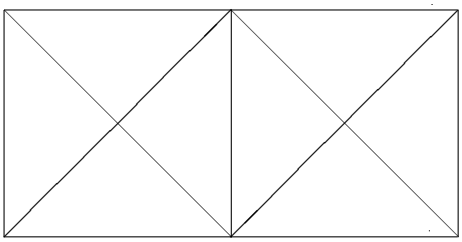
SECTION B : 10 **marks each**

11. Two ladies Cathy and Annet drive between two towns A and B which are 240km apart. Cathy drives from town A at an average speed of 60km/h towards town B. Annet drives from town B at an average speed of 80km/h towards town A.

    (a) Who completes their journey first and why?

    (b) If midway on her journey, Cathy decides to increase her average speed to a level that shall enable her complete the journey at the same time as Annet. What should be her new speed?

12. Count all the triangles in the figure below.



13. Starting at 1 : 00 p.m., Jorge watched three movies. The first movie was 2 hours and 20 minutes long. He took a 20 minute break and then watched the second movie, which was 1 hour and 45 minutes long. He again took a 20 minute break and then watched the last movie, which was 2 hours and 10 minutes long. At what time did the final movie end?

14. A group of friends are sharing a bag of candy.

    On the first day, they eat $\frac{1}{2}$ of the candies in the bag.

    On the second day, they eat $\frac{2}{3}$ of the remaining candies.

    On the third day, they eat $\frac{3}{4}$ of the remaining candies.

    On the fourth day, they eat $\frac{4}{5}$of the remaining candies.

    On the fifth day, they eat $\frac{5}{6}$ of the remaining candies.

    At the end of the fifth day, there is 1 candy remaining in the bag.

    How many candies were in the bag before the first day?

15. Lady Faith is a professional runner. She ran five laps around a track. Her times for the five laps were 65 seconds, 1 minute, 1.5 minutes, 68 seconds, and 57 seconds. What is the average of these times in minutes?

Qn. 1 **Solution :** *Possibilities of angles are* $50°, 65°, 65°$ *and* $50°, 50°, 80°$ *therefore the right pair is D.* ■

Qn. 2 The number of hours in 145 minutes is $\frac{145}{60} = 2$ hours and 25 minutes therefore

$$\begin{array}{rcl} 9 & : & 06 \text{ a.m} \\ 2 & : & 25 \\ \hline 11 & : & 31 \text{ a.m} \end{array}$$

Qn. 3 **Solution :** *The numbers*

$$\frac{5}{6}, \ \frac{6}{7}, \ \frac{2}{5}, \ \frac{1}{3}, \ \frac{4}{5}, \ \frac{3}{7}$$

in percentage will be

$$83\%, 85\%, 40\%, 33\%, 80\%, 43\%$$

so arranging in ascending order we have

$$33\%, 40\%, 43\%, 80\%, 83\%, 85\%$$

in the original fractions we have $\frac{1}{3}, \ \frac{2}{5}, \ \left[\frac{3}{7}, \ \frac{4}{5}\right], \ \frac{5}{6}, \ \frac{6}{7}$ *whose median is*

$$\frac{1}{2}\left(\frac{3}{7} + \frac{4}{5}\right) = \frac{43}{70} = 0.6143$$

Alternative solution

We use the LCM $\{6, 7, 5, 3\} = 210$ *to have the fractions adjusted as*

$$\begin{array}{rcl} \frac{5}{6} \times 210 & = & 175 \\ \frac{6}{7} \times 210 & = & 180 \\ \frac{2}{5} \times 210 & = & 84 \\ \frac{1}{3} \times 210 & = & 70 \\ \frac{4}{5} \times 210 & = & 168 \\ \frac{3}{7} \times 210 & = & 90 \end{array}$$

in ascending order we have $70, 84, [90, 168], 175, 180$ *thus the median is*

$$\frac{90 + 168}{2} = 129$$

which is

$$\frac{129}{210} = \frac{43}{70} = 0.6143$$

■

Qn. 4 **Solution :** *B and D are lions.* ■

Qn. 5 **Solution :** *The area of the rectangle is xy cm^2 and the area of the triangle is* $\frac{1}{2} \times 18 \times x$ *cm^2 thus*

$$xy = \frac{1}{2} \times 18 \times x \quad \Leftrightarrow \quad y = 9cm$$

■

Qn. 6 **Solution :** *The answer is C. Reason since in the column of dogs in the table $y+k_2$ is the largest number. Let x represent the number of cats and y be the number of dogs owned by Coxy and k_1, k_2, k_3 and k_4 are positive numbers then the statements imply,*

	Cats	*Dogs*
Coxy	x	y
Alice	$x - k_1$	$y + k_2$
Bruno	$x - k_3$	$y - k_4$

■

Qn. 7 **Solution :**

$$-3 * -3 = (-3)^2 + 2^{-3} = 9 + \frac{1}{2^3} = 9 + \frac{1}{8} = 9\frac{1}{8}$$

■

Qn. 8 **Solution :** *Let the consecutive numbers be*

$$x,\ x+1,\ x+2,\ x+3,\ x+4,\ \ldots$$

thus the third and the fourth sum is 11. Therefore,

$$(x+2) + (x+3) = 11 \quad \Rightarrow \quad x = 3$$

meaning that the 2022nd term is

$$x + 2021 = 2024.$$

■

Qn. 9 The terms in the sequence will be

$$\begin{aligned} & & 44 \\ 4^2 + 4^2 &= & 32 \\ 3^2 + 2^2 &= & 13 \\ 1^2 + 3^2 &= & 10 \\ 1^2 + 0^2 &= & 1 \\ 1^2 &= & 1 \\ & & 1 \\ & & 1 \\ & & \vdots \end{aligned}$$

thus the 2022nd term is 1.

Qn. 10

$$\begin{aligned} b &= v+5 \\ c &= v-5 \end{aligned}$$

Therefore,

$$b-c=(v+5)-(v-5)=10$$

Qn. 11 (a) **Solution :** *Cathy's time*

$$t=\frac{distance}{speed}=\frac{240}{60}=4hrs$$

Annet's time

$$t=\frac{distance}{speed}=\frac{240}{80}=3hrs.$$

Annet completes first as she takes only 3 *hours and her speed is higher.* ■

(b) **Solution :** *By midway, Cathy will have used* $\frac{120}{60}=2$ *hours. To complete in three hours that Annet uses she must cover the remaining* 120 *km in one hour thus her speed should be* 120 *km/hr.* ■

Qn. 12 The diagram consists of two squares and each square has 8 triangles so the total number of triangles is 16 plus the two triangles formed by combining the two squares giving 18 triangles.

18 triangles	10 marks
17 triangles	8 marks
16 triangles	6 marks
15 triangles	4 marks
14 triangles	1 mark

Qn. 13 **Solution :** *The final movie ended at* 7 : 55 *p.m.*

$$\begin{array}{rr} & 1300 \\ & 0220 \\ & 0020 \\ & 0145 \\ & 0020 \\ + & 0210 \end{array}$$

$$18hr\ (115)minutes = 19hrs\ 55minutes = 1955hrs = 7:55pm$$

■

Qn. 14 **Solution :** *There are many alternative solutions,*

***Solution** 1 At the end* 1 *candy is left, If 1 represent* $\frac{1}{6}$ *on the 5th day so represents* $\frac{1}{6}$ *left at the end of 4th day* $\Rightarrow$ *there were* 6 *left at the end of the 4th day. Since* $\frac{4}{5}$ *are removed on 4th day, there are 6 representing* $\frac{1}{5}$ *left on the 3rd day so there were* $5\times 6=30$ *candies at end of 3rd day. Argueing in this way we get* 720 *candies.*

Solution *2 We use the approach of what is left. Let x be the number of candies in the bag,*

On the first day $\frac{1}{2}$*of* $x = \frac{x}{2}$ *is left.*

On the second day $\frac{1}{3}$*of* $\frac{x}{2} = \frac{x}{6}$ *is left.*

On the third day $\frac{1}{4}$*of* $\frac{x}{6} = \frac{x}{24}$ *is left.*

On the fourth day $\frac{1}{5}$*of* $\frac{x}{24} = \frac{x}{120}$ *is left.*

On the fifth day $\frac{1}{6}$*of* $\frac{x}{120} = \frac{x}{720}$ *is left* $\Leftrightarrow x = 720$ *candies.*

■

Qn. 15 **Solution :** *Let all be expressed in seconds:*

$$65, \quad 60, \quad 90, \quad 68, \quad 57.$$

The average times is $\frac{340}{5} = 68$ *seconds which is an equivalent to* $1\frac{8}{60}$ *minutes.* ■